

Combining Virtual Reality with Camera Data and a Wearable Sensor Jacket to Facilitate Robot Teleoperation

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ABSTRACT

Unmanned ground vehicles (UGV) with differing degrees of autonomy are increasingly used for routine tasks, albeit still widely controlled through teleoperation in safety-critical contexts. We propose to complement a traditional approach of showing camera views on a 2D screen with an immersive Virtual Environment (VE), displaying camera data with depth cues around a model of the robot. Combining both approaches with a wearable sensor jacket, the operator can choose the best one for the current task. Preliminary experiments showed that even untrained operators can successfully solve pick and place tasks with high accuracy with our system.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Computer systems organization—Embedded and cyber-physical systems—Robotics—External interfaces for robotics

1 INTRODUCTION

Using UGVs is ideal for any situation posing a risk to human personnel, like in Explosive Ordnance Disposal (EOD) or radiological and nuclear inspection tasks. These robots are typically controlled through manual teleoperation. Downsides are complex robot controls and the need for highly trained operators. Also, using a UGV in the field is a highly stressful task for such an operator. Therefore, we provide an intuitive Virtual Reality (VR) interface for dexterous tasks such as telemanipulation. We try to enhance perception and situational awareness for the operator while keeping the cognitive burden, stress level, and symptoms of Simulator Sickness (SS) low. Main aspects of our system are a high level of immersion for the remote operator, an intuitive way of controlling a UGV with a mounted manipulator, and providing semi-autonomous assistance functions.

2 PREVIOUS AND RELATED WORK

We previously conducted an extensive user study to compare two different immersive teleoperation systems, with directly coupled stereoscopic camera view and head-mounted display (HMD), with a traditional 2D screen approach [3]. This study included three different tasks with differing complexity and 19 participants, including 4 EOD experts. Using a within-subjects design we examined 171 datasets and concluded that each system had its specific advantages and disadvantages. The measurable benefit of using an HMD to show stereoscopic camera data was highly task-dependant and sometimes negated by side-effects, such as up to severe symptoms of SS. We concluded, that the best solution might be a combination of both approaches, swapping on a case-by-case basis for each task.

Using VR hardware for teleoperation is not a new concept. Especially the last years have seen several proposals from others, like [5].

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Figure 1: Four of the mounted cameras on the robot are made available in the Virtual Environment (VE). Two of them are RGB-D cameras and deliver depth information in the VE, the other two are monoscopic cameras, which are mapped to a virtual 2D screen.

We specifically chose to not fully rely on VR hardware though, to have the VE as optional drop-in solution available.

3 SYSTEM DESCRIPTION

The UGV exemplary used in this work is a modified, tracked Telerob tEODor robot with a mounted 7-DOF Telerob telemax manipulator with a parallel gripper. Both parts of the hardware are current, specialized EOD systems. Therefore, communication between robot and base station is typically realized with a radio link with limitations on bandwidth and update rate. The UGV is fitted with four cameras used by our proposed system: For overview an RGB-D camera (Asus Xtion Pro) attached to a SCHUNK PW-70 pan-tilt unit on the back of the robot, a second RGB-D camera (Intel RealSense D435) providing a highly detailed view of the surroundings of the gripper, and two basic analog cameras from the robot manufacturer, mounted along the manipulator. All camera positions are depicted in Fig. 1.

From a technical side, we chose the robot framework ROS (Robot Operating System), running on the robot and a Linux laptop as control station. A second laptop running Windows was used for the VR side of the system. A visualization and enhanced control module was realized with the 3D game engine Unity3D. In Unity3D, we also used SteamVR to interact with the VR hardware, an HTC Valve with one controller. Communication from the robot to Unity3D was done via Rosbridge and WebSockets, based on ROS Reality [2].

For controlling the manipulator, we offer an intuitive wearable sensor jacket, which works with four inertial measurement units (IMU). The jacket allows direct control of the manipulator's tip by constantly sending goal poses, while the manipulator configures itself to try and reach those accurately. Using the jacket has been shown to outperform traditional joystick control of a manipulator [1], is robust to failures, such as momentary tracking errors, and is easily exchangeable between operators on-the-fly. For integrating the sensor jacket into our system, we updated it to use wireless IMUs. The operator can start and stop the direct coupling by using the trigger on the VR controller or, as fallback, with a cabled control stick.

Instead of directly linking the live video feed of an overview stereo camera system to the HMD, we decided to decouple the user's view. Therefore, we created a VE (Fig. 2), in which we show a virtual robot model which updates to the current configuration with 10 Hz. Furthermore, we link the two mounted RGB-D cameras to their virtual counterpart. Thus, we can use depth images, transmitted with 25 Hz, and a custom shader to project a 3D representation of the current camera view into the VE. Even though the operator can not directly move a camera through their head movement anymore, we believe that being able to freely roam the VE and having a very dynamic view of the scene can deliver an equally immersive experience. To move the overview camera, the operator can use the touchpad on the VR controller. Additionally, we link the two monoscopic cameras to the VR controller and alternately one of them can be shown on a virtual 2D screen. Altogether, we see several advantages of this concept. Main point is the decoupled view shown inside the HMD, which can be updated with the frame rate of the display during head movements. Previously limiting factors were communication and processing delays and mechanical latency. Other advantages are the parallel display of several available camera feeds and the possibility of free movement in the VE. Especially if something is shown further away from the UGV details can be hard to spot. The user can now inspect the shown scene from any angle or distance. Especially for manipulation tasks or navigation in close quarters this can have a big impact on task success. For ease of use, we also implemented an exemplary assistance function, to be triggered with the VR controller, which semi-autonomously moves the manipulator into a fixed pose.

Please refer to the accompanying video for a short demonstration of the VR system with the sensor jacket (<https://youtu.be/9c4p29mzR6M>).

4 PRELIMINARY RESULTS

To get a first impression of the usefulness of our proposed system in the field, we used an early prototype to successfully participate at the 2019 EnRicH robot competition [4]. This event took place in the nuclear reactor Zwentendorf in Austria and involved three challenging tasks for the teams, including navigation and manipulation with a high amount of situational awareness needed. Especially the reactor-typical obstacles, like barriers in low height, stairs, ramps, and narrow passages, increased the difficulty for a robot operator. Due to the limited trial time, we chose a trained operator for the competition and were able to showcase the advantages of our proposed hybrid system with VR support to an international jury.

Another chance to gather preliminary results provided our interactive booth at the conference "Futuras in Res" end of 2019, where we ran a live demonstration of the VR system, see Fig. 3. Having incorporated valuable feedback from the preceding robot competition our system was running straight for two days, ten hours each, allowing to swap operators simply by exchanging the sensor jacket and VR controller. Roughly 50 of the visitors, with a wide variety of technical background from science, industry and politics, volunteered to participate in the demonstration as novice operators. All users were given a few-minute introduction to the system and briefed for the manipulation task, a pick and place operation. The picking part was fully teleoperated, whereas the following place operation could be run semi-autonomously. We provided an assistance function for that part, which would at the press of a controller button generate a collision-free path to the place location and would be executed under supervision of the user. They could then manually drop the grasped object into the target bin. Everyone involved was able to safely and successfully solve the given task. Albeit the short exposure to the VR system, with around 15 minutes per participant, we received no feedback of noticeable amounts of SS. On the contrary, many of the volunteers made remarks especially on the intuitive and easy usage of the system.



Figure 2: In the VE, we show an approximated robot model, standing on a virtual floor. Also visible here is the output of the RGB-D cameras, which augments the virtual scene with real-world data.



Figure 3: A prototype of the whole system was showcased at the fair of the conference "Futuras in Res" in Berlin, Nov. 2019. Visitors were given a short introduction and could then try the manipulation task.

5 CONCLUSION AND FUTURE WORK

We proposed a new, hybrid concept for teleoperation of a UGV, using an optional, free-to-roam VE scene with a virtual robot model in its current configuration, augmented with sensor data from mounted cameras. Furthermore, we described the integration of an intuitive control method via a wearable sensor jacket, which is also usable outside the VE. First results show a promising reduction of SS symptoms while volunteers successfully tried the system on pick and place tasks. The proposed system could facilitate the teleoperation of a robot so that even novice users could use the system within minutes of briefing.

Because of the current worldwide situation we could not yet run a systematic evaluation of the current system with an adequate amount of participants. Therefore, this will be the next step for future work. We are planning to run an experiment with different tasks, comparing the proposed VR system against the traditional 2D screen approach.

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